UNIVERITY OF RAJASTHAN

JAIPUR-302004

THREE/FOUR-YEAR UNDERGRADUATE PROGRAMME

Name of Faculty	**Science**
Name of Discipline	**Chemistry**
Type of Discipline	**Major**
List of Programme offered as Minor Discipline	**-NA-**
Offered to Non-Collegiate Students	**-Yes**

Programme: UG0802/03 – Three/Four Year Bachelor of Science

I & II Semester

(Syllabus as per NEP-2020 and Choice Based Credit System)

(Academic Year 2025-26 onwards)

SEMESTER-WISE PAPER TITLES WITH DETAILS

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UG0802/03 – Three/Four Year Bachelor of Science								
S. No.	Level	Semester	Type	Chemistry	Credits			
				Course Title	L	T	P	Total
1.	5	I	MJR	UG0802/03 – CHM-51T-101 – Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.	4	0	0	4
2.	5	I	MJR	UG0802/03– CHM-51P-102 – Practical I	0	0	2	2
3.	5	II	MJR	UG0802/03 – CHM-52T-103 – Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.	4	0	0	4
4.	5	II	MJR	UG0802/03 – CHM-52P-104 – Practical II	0	0	2	2
5.	6	III	MJR	UG0802/03 – CHM-63T-201 –	4	0	0	4
6.	6	III	MJR	UG0802/03 – CHM-63P-202 – Practical III	0	0	2	2
7.	6	IV	MJR	UG0802/03 – CHM-64T-203 –	4	0	0	4
8.	6	IV	MJR	UG0802/03 – CHM-64P-204 – Practical IV	0	0	2	2
9.	7	V	MJR	UG0802/03 – CHM-75T-301 --Hard & Soft Acids and Bases, Transition metal complexes, Spectroscopy, Organosulphur compounds, Synthetic Polymers, Drugs & Dyes. Electrochemistry.	4	0	0	4
10.	7	V	MJR	UG0802/03 – CHM-75P-302 – Practical V	0	0	2	2
11.	7	VI	MJR	UG0802/03 –CHM-76T-303 – Bioinorganic chemistry, Organometallic chemistry, Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum mechanics and MOT	4	0	0	4
12.	7	VI	MJR	UG0802/03 – CHM-76P-304 – Practical VI	0	0	2	2
13.	8	VII	MJR	UG0802/03 – CHM-87T-401 –	4	0	0	4
14.	8	VII	MJR	UG0802/03 – CHM-87P-402 – Practical VII	0	0	2	2
15.	8	VIII	MJR	UG0802/03 – CHM-88T-403 –	4	0	0	4
16.	8	VIII	MJR	UG0802/03 – CHM-88P-404 – Practical VIII	0	0	2	2

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PROGRAMME PREREQUISITES/ELIGIBILITY
12th standard pass in science from CBSE, RBSE or a recognized board of education.
PROGRAMME OUTCOMES (POs)
1. **Conceptual knowledge of chemical science:** Students will get acquainted with the conceptual knowledge of chemical science which will help them to understand the subject and it will be beneficial in long run. 2. **Training to manage unusual and unexpected incidents/disasters:** The knowledge of chemical science will help them to deal with unusual incidents in the neighborhood. Sudden explosion by chemicals and excessive misuse of unwanted substances can be managed with basic knowledge of chemistry as well as environmental pollution may be controlled by the students by spreading awareness in the society about the harmful pollutants viz; plastic, pesticides, harmful smog, unused drugs etc. 3. **Laboratory Experimental Skills:** As we know the fact that trials are an essential part of an exploration in our life therefore the students will gain practical experience by conducting experiments, using laboratory instruments and apparatus. 4. **Employment opportunities:** Students will acquire employment in the various national and private R & D sectors such as: The students with the strong chemistry background can get jobs in chemical and related industries viz. Agrochemicals, Metallurgical, Fertilizer, Biofertilizer, Textile, Food, Ceramics, Cement, Petrochemicals, Pesticides, Plastics, Polymers, etc. The students can find opportunities in Pharmaceutical companies, ForensiLab,etc. Petroleum, Soil Testing Labs, Environment consulting firms and other sectors such as Analytical Chemist, Chemical Product Officer, Radiologist and Toxicologist. 5. **Integrated M. Sc-Ph.D courses at prestigious institutions:** After completing this bachelor's degree course, students can get engaged in integrated M,Sc- Ph D courses or can get Master's degree in various interdisciplinary fields at prestigious institutions like CSIR, IISc, IITs, NCL (national chemical laboratories), IISERs, NISER etc.

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Examination Scheme:

- **1 Credit = 25 marks for examination/evaluation.**
- **For Regular Students:**
  - There will be Continuous assessment, in which sessional work and the terminal examination will contribute to the final grade. Each course in Semester Grade Point Average (SGPA) has two components - Continuous assessment (20% weightage) and (**End of Semester Examination**) EoSE (80% weightage).
  - 75% Attendance is mandatory for appearing in EoSE.
  - To appear in the EoSE examination of a course/subject student must appear in the mid-semester examination and obtain at least a C grade in the course/subject.
  - Credit points in a Course/Subject will be assigned only if, the regular student obtains at least a C grade in the **CA (Continuous Assessment)** and EoSE examination of a Course/Subject.

- **In case of the Non-Collegiate Students**:

  - There will be no Continuous Assessment and credit points in a Course/Subject will be assigned only if, the Non-Collegiate Student obtains at least a C grade in the EoSE examination of a Course/Subject.

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Examination scheme for Continuous assessment (CA)

DISTRIBUTION OF CONTINUOUS ASSESSMENT (CA) MARKS

S. No.	CATEGORY	Weightage (out of total internal marks)		THEORY					PRACTICAL		
				CORE (Only Theory)	CORE (Theory + Practical)	AEC	SEC	VAC	CORE (Theory + Practical)	SEC	VAC
	Max Internal Marks			30	20	20	10	10	10	10	10
1	Mid-term Exam	50%		15	10	10	5	5	5	5	5
2	Assignment	25%		7.5	5	5	5	2.5	2.5	2.5	2.5
3	Attendance	25%		7.5	5	5	5	2.5	2.5	2.5	2.5
		Regular Class Attendance	= 75%	3	2	2	1	1	1	1	1
			75 – 80%	4	3	3	1.5	1.5	1.5	1.5	1.5
			80 – 85%	5	4	4	2	2	2	2	2
			> 85%	7.5	5	5	2.5	2.5	2.5	2.5	2.5

Note:

1. Continuous assessment will be the sole responsibility of the teacher concerned (under the heading assignment, interactive sessions/ group discussion among students may be conducted by the concerned teacher / PPT for selective topics may be assigned by the teacher at college level).
2. For continuous assessment no remuneration will be paid for paper setting, Evaluation, Invigilation etc.
3. For continuous assessment Paper setting and Evaluation responsibility will be of teacher concern.
4. For continuous assessment no Answer sheets/question papers etc. will be provided by the University.
5. Colleges are advised to keep records of continuous assessment, attendance etc.

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I – Semester

Examination Scheme for EoSE

- **CA – Continuous Assessment**
- **EoSE – End of Semester Examination**

For Regular Students

Type of Examination	Course Code / Nomenclature	Duration of Examination		Maximum Marks		Minimum Marks	
Theory	CHM-51T-101 – Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.	CA	1 Hr.	CA	20	CA	8
		EoSE	3 Hrs.	EoSE	80	EoSE	32
Practical	CHM-51P-102 – Practical I	CA	1 Hr.	CA	10	CA	4
		EoSE	4 Hrs.	EoSE	40	EoSE	16

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B.

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – B: 60 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 15 marks

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For Non-Collegiate Students –

Type of Examination	Course Code and Nomenclature	Duration of Examination	Maximum Marks	Minimum Marks
Theory	CHM-51T-101 - Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.	3 Hrs.	100	40
Practical	CHM-51P-102 – Practical I	4 Hrs.	50	20

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – A: 80 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 20 marks.

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Syllabus

I – Semester – [Chemistry]

[UG0802/03]-[CHM-51T-101]- **Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.**

[UG0802/03]-[CHM-51P-102]- **PRACTICAL – I**

<table>
<tr><th>Semester</th><th>Code of the Course</th><th colspan="3">Title of the Course/Paper</th><th colspan="2">NHEQF Level</th><th>Credits</th></tr>
<tr><td>I</td><td>CHM-51T-101</td><td colspan="3">Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.</td><td colspan="2">5</td><td>4</td></tr>
<tr><td>I</td><td>CHM-51P-102</td><td colspan="3">PRACTICAL-I</td><td colspan="2">5</td><td>2</td></tr>
<tr><td rowspan="2">Level of Course</td><td rowspan="2">Type of the Course</td><td colspan="3">Credit Distribution</td><td rowspan="2">Offered to NC Students</td><td colspan="2" rowspan="2">Course Delivery Method</td></tr>
<tr><td>Theory</td><td>Practical</td><td>Total</td></tr>
<tr><td>5</td><td>Major</td><td>4</td><td>2</td><td>6</td><td>Yes</td><td>Through Lecture, Sixty (60) Lectures</td><td>Class room Teaching/Power-Point (PPT)</td></tr>
<tr><td colspan="2">List of Programme Codes in which offered as Minor Discipline</td><td colspan="6">-NA -</td></tr>
<tr><td colspan="2">Prerequisites/Eligibility</td><td colspan="6">12th standard pass in science from CBSE, RBSE or a recognized board of education.</td></tr>
<tr><td colspan="2">Course Objectives:</td><td colspan="6">The main aim of this course is to provide students with a theoretical and conceptual understanding of s-, p-block elements chemistry, noble gases, nuclear chemistry and the fundamental concepts of organic chemistry with their reaction mechanisms, generation and stability of various intermediates, including stereochemistry. The objective of this course is to understand the basic concepts of mathematics and to get knowledge in the chemical kinetics of reactions. Moreover, the laboratory course is further designed to provide students with practical experience in basic qualitative analytical techniques including basic laboratory techniques.</td></tr>
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Detailed Syllabus

Semester-I – [Chemistry]

UG0802/03 CHM-51T-101 (4 Hrs./week)	**Chemistry of s & p-block elements. Noble Gases, Nuclear Chemistry, Fundamentals of Organic Chemistry, Stereochemistry, Mathematical Concepts and Chemical Kinetics.**

Unit-I

Periodicity of s and p-block elements:
Effective nuclear charge, Shielding or screening effect, Slater rules, Variation of effective nuclear charge in periodic table. Atomic radii (van der Waals), Ionic and crystal radii, Covalent radii (octahedral and tetrahedral). Ionization enthalpy, Successive ionization enthalpies and factors affecting ionization energy. Applications of ionization enthalpy. Electron gain enthalpy, Trends of electron gain enthalpy. Electronegativity, Pauling's/Mulliken's/Allred-Rochow's and Mulliken-Jaffé's electronegativity scales. Variation of electronegativity with bond order, Partial charge, Hybridization, Group electronegativity. Sanderson's electron density ratio.

Periodicity in properties of p-block elements with special reference to atomic and ionic radii, Ionization energy, Electron-affinity, Electronegativity, Diagonal relationship, Catenation.

s-Block Elements: Comparative study of properties of alkali and alkaline earth metals, Diagonal relationships, Salient features of hydrides, Solvation and complexation tendencies including their functions in biosystems, An introduction to alkyls and aryls.

Unit-II

Some Important Compounds of p-block Elements: Hydrides of boron, Diborane and higher boranes, Borazine, Borohydrides, Fullerenes, Carbides, Fluorocarbons, Silicates (structural principle), Tetrasulphurtetranitride, Basic properties of halogens, Interhalogens and polyhalides.

Chemistry of Noble Gases: Chemical properties of the noble gases, Chemistry of Xenon, Structure and bonding in Xenon compounds.

Nuclear Chemistry: Fundamental particles of nucleus (nucleons), concept of nuclides and its representation, Isotopes, Isobars and Isotones (with specific examples), forces operating between nucleons (n-n, p-p & n-p), Qualitative idea of stability of nucleus (n/p ratio).
Radiochemistry: Natural and artificial radioactivity, Radioactive disintegration series, Radioactive displacement law, Radioactivity decay rates, Half-life and average life, Nuclear binding energy, Mass defect and calculation of defect and binding energy, Nuclear reactions, Spallation, Nuclear fission and fusion. Brief discussion on atom bomb, Nuclear reactor and Hydrogen atom.

15 Lectures

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Unit-III

Fundamentals of organic chemistry: Organic Compounds: Classification and Nomenclature, Hybridization, Shapes of molecules, Influence of hybridization on bond properties. Electronic Displacements: Inductive, Electromeric, Resonance and mesomeric effects, Hyperconjugation and their applications; Dipole moment; Organic acids and bases; their relative strength. Homolytic and Heterolytic bond fission with suitable examples. Curly arrow rules, Formal charges; Electrophiles and Nucleophiles; Nucleophlicity and basicity.

Stereochemistry of Organic Compounds: Concept of isomerism, Types of isomerism, Difference between configuration and conformation, Flying wedge and Fischer projection formulae.

Optical Isomerism: Elements of symmetry, Molecular chirality, Enantiomers, Stereogenic centre, Optical activity. Properties of enantiomers, Chiral and achiral molecules with two stereogenic centres. Diastereomers, Threo and erythro isomers, Meso compounds. Resolution of enantiomers. Inversion, Retention and racemization (with examples).

Relative and absolute configuration, Sequence rules, D/L and R/S systems of nomenclature.

Geometrical Isomerism: Determination of configuration of geometric isomers - cis/trans and E/Z systems of nomenclature. Geometrical isomerism in oximes and alicyclic compounds.

Conformational Isomerism: Newman projection and Sawhorse formulae, Conformational analysis of ethane, n-butane and cyclohexane.

Unit-IV

Mathematical Concepts:

Logarithmic relations, Curve sketching, Linear graphs and calculations of slopes, Differentiation of functions like: kx, e^{x}, x^{n}, $\sin x$ and $\log x$. maxima and minima, Partial differentiation and Euler's reciprocity relations, Integration of some useful/relevant functions. Permutations and combinations, Factorials, Probability, Matrices and Determinant.

Chemical Kinetics:

Chemical kinetics and its scope, Rate of a reaction, Factors influencing the rate of a reaction: concentration, temperature, pressure, solvent, light, catalyst. Concentration dependence of rates, Mathematical characteristics of simple chemical reactions - zero order, first order, second order and pseudo-order; Half-life and mean-life. Determination of the order of reaction - differential method, Method of integration, Method of half-life period and isolation method.

Radioactive decay as a first order phenomenon.

Experimental methods of chemical kinetics: Conductometric, Potentiometric, Optical methods: polarimetry and spectrophotometric method. Theories of chemical kinetics. Effect of temperature on rate of reaction, Arrhenius equation, Concept of activation energy.

Simple collision theory based on hard sphere model transition state theory (equilibrium hypothesis). Expression for the rate constant based on equilibrium constant and thermodynamic aspects.

15 Lectures

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Suggested Books and References:

1. Concise Inorganic Chemistry by J.D. Lee, Wiley.
2. Inorganic Chemistry by Catherine E. Housecroft and Alan G. Sharpe, Pearson.
3. Selected Topics in Inorganic Chemistry by Wahid U. Malik, G. D. Tuli and R. D. Madan, S. Chand, New Delhi.
4. Advanced Inorganic Chemistry: Volume I & II by Satya Prakash, G. D. Tuli, S. K. Basu and R. D. Madan, S. Chand, New Delhi.
5. Inorganic Solids – Introduction to Concepts in Solid-state Structural Chemistry by D. M. Adams, John Wiley, London.
6. Principles of Inorganic Chemistry by Puri, Sharma & Kalia, Vishal Publishing Co.
7. Essentials of Nuclear Chemistry by H.J. Arnikar, New Age International Publishers.
8. Organic Chemistry by I. L. Finar, Pearson.
9. Organic Chemistry by R.T. Morrison, R.N. Boyd & S.K. Bhattacharjee, Pearson.
10. Stereochemistry conformation and Mechanism by P.S. Kalsi, New Age International Publishers.
11. Stereochemistry of Organic Compounds by V. K. Ahluwalia, Springer.
12. Chemical Kinetics by Keith J. Laidler, Pearson Education.
13. Principles of Physical Chemistry by B. R. Puri, L. R. Sharma & M. S. Pathania, Vishal Publishing Co.
14. Advanced Physical Chemistry by Gurdeep Raj, Goel Publishing House.
15. Physical Chemistry by W. Atkins, Oxford University Press.
16. Physical Chemistry by R. J. Silby and R. A. Alberty, John Wiley & Sons.
17. Physical Chemistry by G.M. Barrow, Tata McGraw-Hill.
18. A Textbook of Physical Chemistry: (Volume I) by K. L. Kapoor, Macmillan India Ltd.

Suggested E-resources:

All the above suggested books are available as e- books.

Online Lecture Notes and Course Materials:

All prescribed courses are available in the form of e-books, Adobe Acrobat documents (PDF), web pages etc.

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SYLLABUS

CHM-51P-102: Practical – I (4 Hrs./week)

Inorganic Chemistry **10 marks**

1. **Volumetric Analysis**
   (a) Estimation of hardness of water by EDTA.
   (b) Estimation of ferrous/ferric ions by dichromate/permanganate method.
   (c) Estimation of copper using thiosulphate by iodometric method.
   (d) Determination of quantity of acetic acid in commercial vinegar using standard NaOH solution.
   (e) Determination of alkali content in antacid tablet using standard HCl solution.
   (f) Estimation of calcium content in chalk as calcium oxalate by permanganometery.

Organic Chemistry **10 marks**

2. **A. Laboratory Techniques** **3 marks**
   (a) Determination of melting point (naphthalene, benzoic acid, urea etc.); boiling point (methanol, ethanol, cyclohexane, etc.); mixed melting point (urea-cinnamic acid) using Thiele's tube.
   (b) Crystallization of phthalic acid and benzoic acid from hot water, acetanilide from boiling water, naphthalene from ethanol. Sublimation of naphthalene, camphor etc.

   B. Qualitative Analysis **7 marks**
   Identification of functional groups (unsaturation, phenolic, alcoholic, carboxylic, carbonyl, ester, carbohydrate, amine, amide, nitro, etc.) in simple organic compounds (solids or liquids) through element detection (N, S and halogens).

Physical Chemistry **10 marks**

3. **Viscosity and Surface Tension:**
   (a) To determine the viscosity/surface tension of a pure liquid (alcohol etc.) at room temperature. (Using the Ostwald Viscometer/Stalagmometer, respectively.).
   (b) To determine the percentage composition of a given binary mixture (acetone and ethyl methyl ketone) by surface tension method.
   (c) To determine the percentage composition of a given binary mixture (non- interacting systems) by viscosity method.
   (d) To determine the viscosity of amyl alcohol in water at different concentrations (hange in viscosity due to intermolecular interactions and calculate the excess viscosity of these solutions.

4. **Viva voce** **5 marks**

5. **Practical Record** **5 marks**

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Suggested Books and References:

1. Vogel's Qualitative Inorganic Analysis, A. I. Vogel, Prentice Hall.
2. Vogel's Quantitative Inorganic Analysis Including Elementary Instrumental Analysis, ELBS.
3. Vogel's Textbook of Quantitative Chemical Analysis, A. I. Vogel, Pearson Education Ltd.
4. Advanced Practical Organic Chemistry by N. K. Vishnoi, Vikas Publishing House Pvt Ltd.
5. Comprehensive Practical Organic Chemistry: Preparation and Quantitative Analysis, V. K Ahluwalia. Universities Press, Hyderabad.
6. Laboratory Techniques in Organic Chemistry by V. K Ahluwalia, I K International, N
7. Advanced Practical Organic Chemistry J. B Yadav, Goel Publishing House.
8. Practical Physical Chemistry, by B. D Khosla, S. Chand & Company.

Suggested E-resources:

All the above suggested books are available as e- books.

Online Lecture Notes and Course Materials:

All prescribed courses are available in digital form in the form of e-books, Adobe Acrobat documents (PDF), web pages etc.

Course Learning Outcomes:

By the end of this course, students will acquire clear understanding of various concepts related to chemistry of s & p-block elements, noble gases, nuclear chemistry, fundamentals of organic chemistry, classifying the molecules as chiral or achiral, determining the D/L and R/S nomenclature of stereoisomers and identifying the formation of racemic mixture or optically active compounds, mathematical concepts along with chemical kinetics of reactions. Students will also have practical experience in basic laboratory techniques including determination of various physical properties of substances, crystallization and preparation of standard solutions of different concentrations along with identification of functional groups in organic compounds,.

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